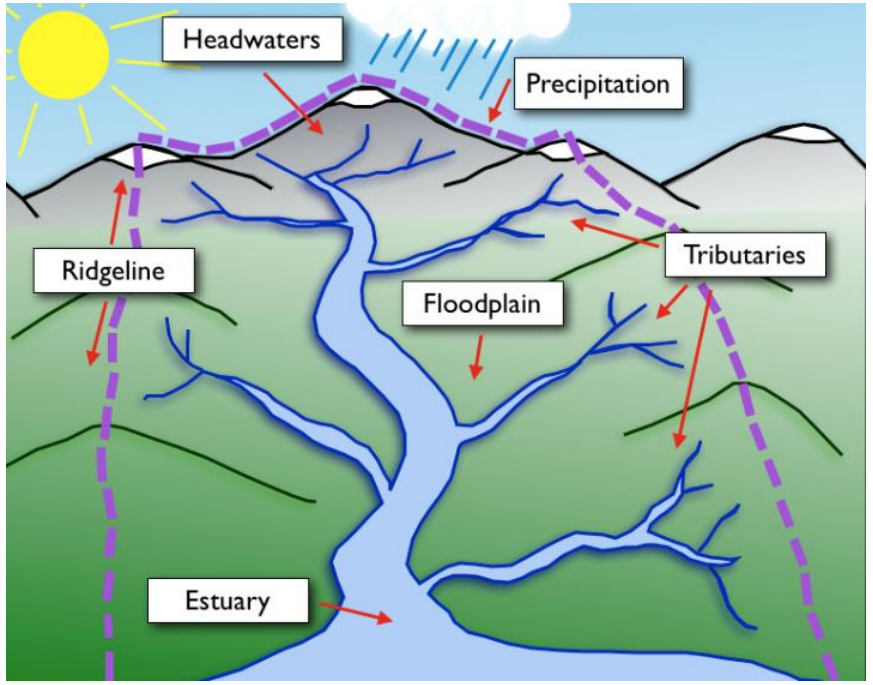
Hydrological Modeling

Hydroinformatics
US Pakistan Center for Advanced Studies in Water
MUET, Jamshoro
Fall, 2026



Learning Objectives

1. Explain how rainfall is summarized and transformed into effective precipitation.
2. Describe watershed boundaries, drainage networks, and key basin characteristics.
3. Understand the water budget and basic rainfall–runoff relationship.
4. Explain components of a hydrograph and how catchment and storm characteristics influence runoff response.
5. Connect these concepts to rainfall–runoff modeling.

1. Rainfall Analysis

From measured rainfall to rainfall available for runoff

Mean Rainfall

- Mean annual rainfall: average annual rainfall over a long period; roughly 30–40 years of records.
- Mean monthly rainfall: average rainfall for a given month over multiple years.
- Comparing observed annual rainfall with the long-term mean helps identify years with rainfall deficit or surplus.
- Use a sufficiently long and continuous record when characterizing climatological rainfall.

Rainfall Characteristics

- Rainfall varies in time and space, so a single gauge may not represent an entire catchment.
- Storm **intensity**, **duration**, and **spatial distribution** are especially important for runoff generation.
- Rainfall analysis provides the input information for rainfall–runoff analysis.

From Gross Rainfall to Effective Rainfall



Effective precipitation = gross precipitation − losses

- Effective rainfall is the portion of rainfall that produces a perceptible streamflow response.
- Not all rainfall becomes surface runoff; some infiltrates, is stored, or contributes to subsurface flow.

Infiltration and Runoff Generation

- Dry soil may initially have a high infiltration capacity.
- If rainfall intensity/rate is below infiltration capacity, little or no overland flow occurs, although interflow may still develop.
- With continuing rainfall, soil becomes wetter, and infiltration capacity can decline.
- When rainfall exceeds the capacity of the soil to absorb water, excess water contributes to runoff.

2. Watershed Concepts

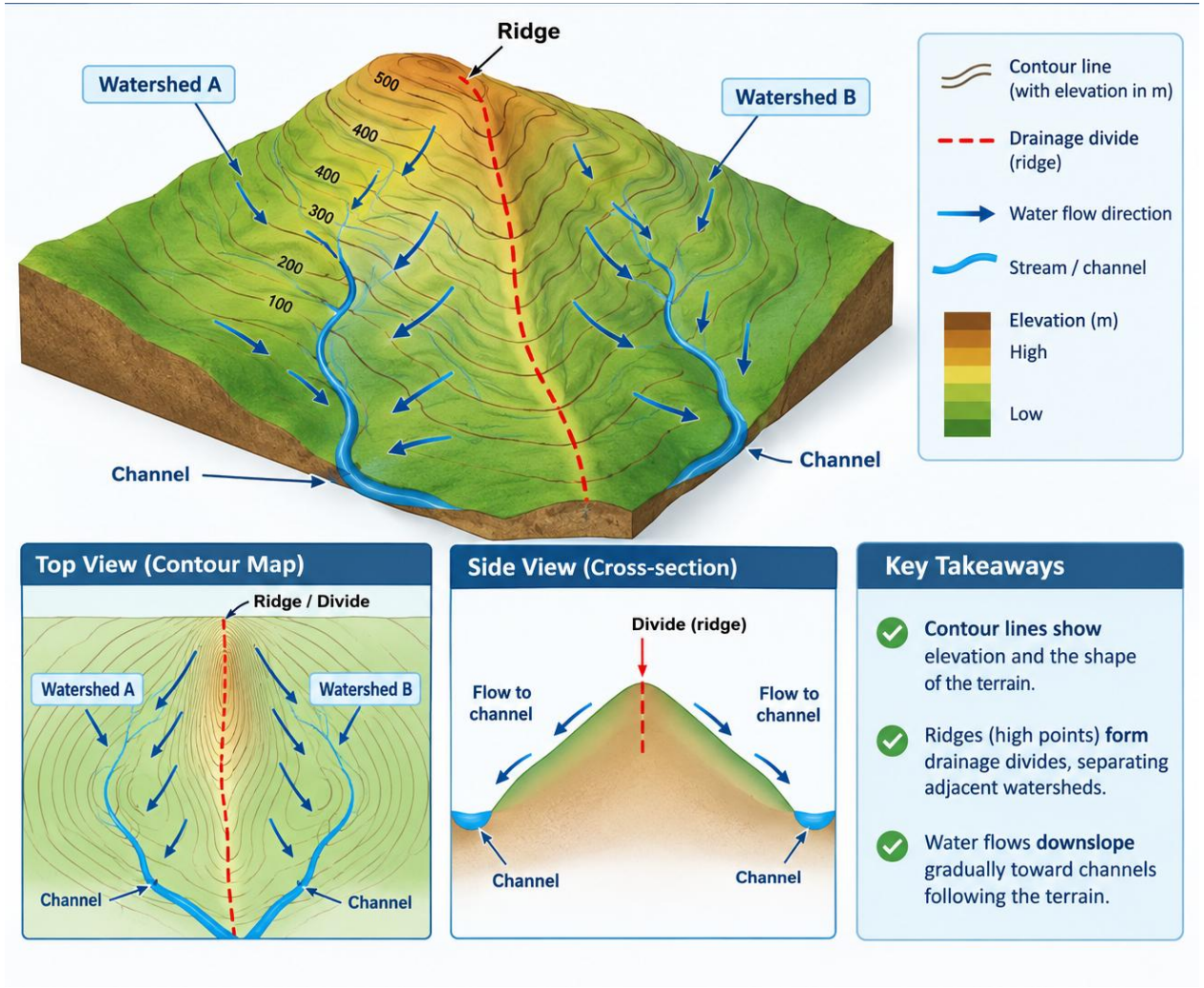
The watershed is the basic spatial unit for hydrologic analysis

What is a watershed?

- A watershed is the area that contributes flow to a specified point or outlet.
- It is a basic hydrologic unit within which measurements, calculations, and predictions are made.
- Watersheds can be very small or can encompass major river basins such as the Indus Basin.
- Boundaries are commonly defined by topographic divides or ridgelines.

Reading a Watershed from Topography

- Use contour lines and ridges to identify drainage divides.
- Water flows downslope toward channels.



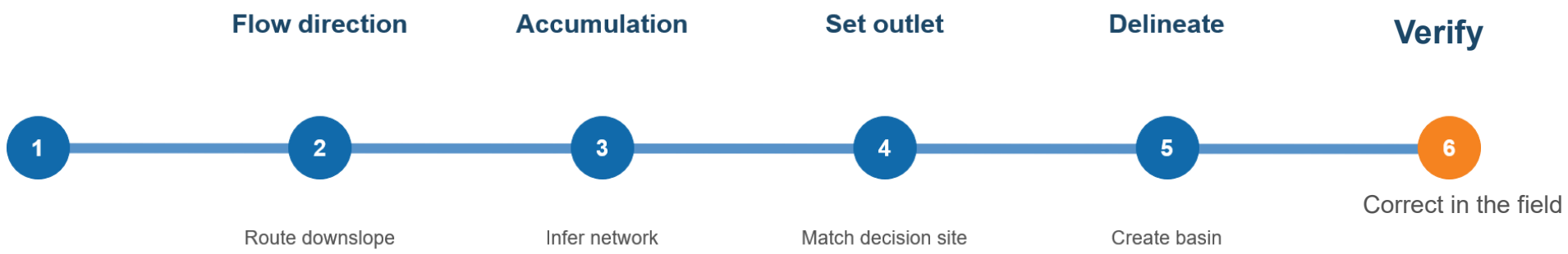
Watershed Delineation: The Core Idea

1. Select the outlet or pour point.
2. Identify the surrounding topographic divide.
3. Trace the boundary so that all land inside drains toward the outlet.
4. The resulting polygon represents the contributing watershed.
5. GIS watershed delineation automates this logic using a DEM and flow-direction/flow-accumulation analysis.

From DEM to Watershed Boundary

A computer-generated boundary should be checked against the landscape.

- Check the DEM and correct important errors.
- Identify flow direction and the stream network.
- Set the outlet at the actual decision site.
- Delineate the watershed and calculate its area.
- Verify channels and drainage paths in the field.



What a DEM can—and cannot—tell you

01 Reliable uses

Delineation, slope, flow direction, relative relief.

02 Common errors

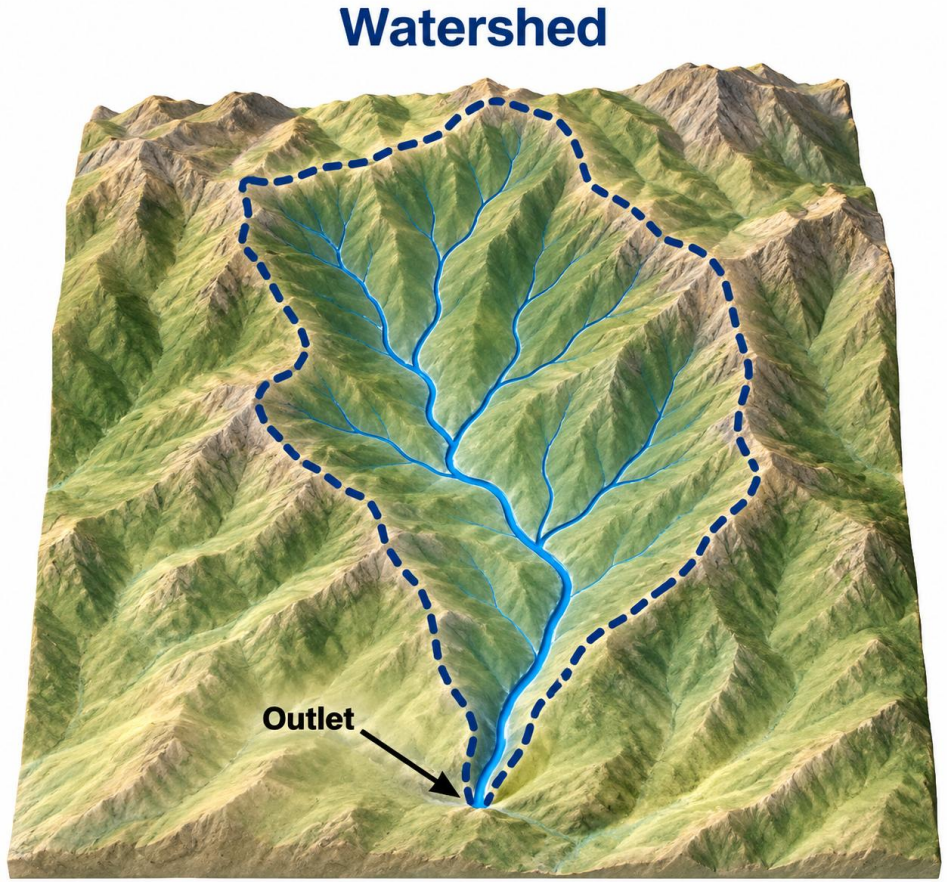
Road embankments, flat areas, sinks, and poorly represented channels.

03 Field checks

Outlet position, channel continuity, and cross-drainage.

04 Available sources

SRTM and other DEMs; document resolution and date.



FROM MAP TO FIELD

Drainage Network

1. A drainage network is the set of cells or channels through which surface water flows.

2. Flow accumulation identifies locations receiving flow from many upstream cells.

3. A stream can be defined by applying a contributing-area threshold.

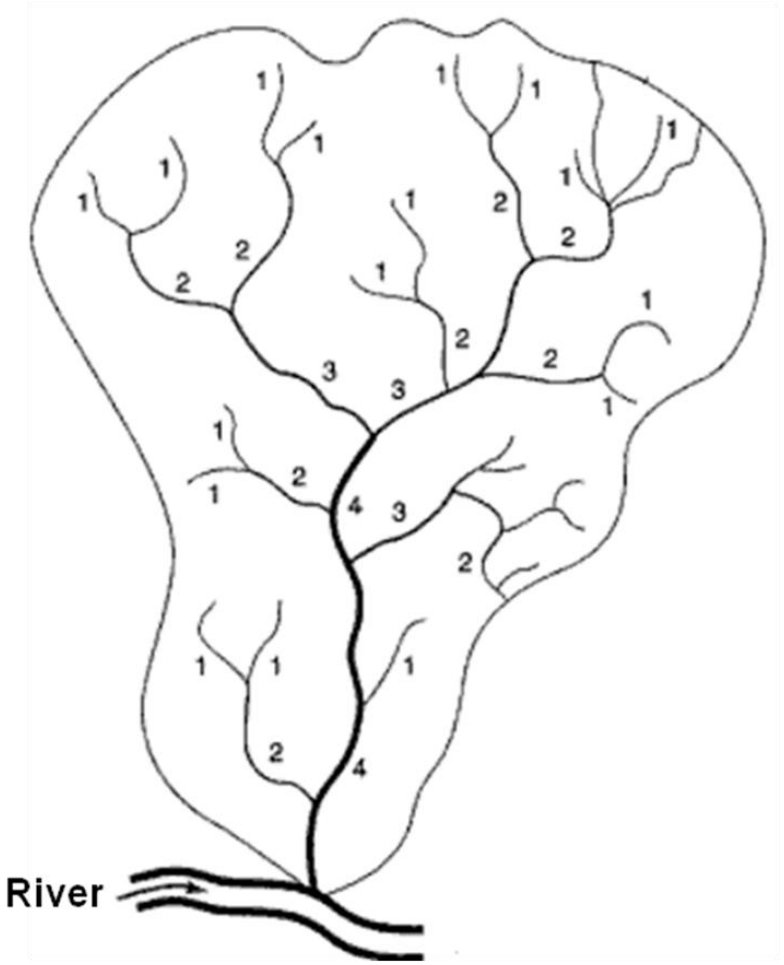
4. The threshold should reflect the scale and purpose of the analysis; it is not a universal value.

Key Basin Characteristics

- Drainage area — area contributing flow to a common outlet.
- Drainage divide — boundary separating watersheds draining to different outlets.
- Drainage density — total stream length per unit basin area.
- Stream frequency — number of channels per unit area.
- Stream gradient — elevation drop per unit channel length.
- Basin relief — highest elevation minus lowest elevation.
- Time of concentration — travel time from the hydraulically most distant point to the outlet.

Stream order — Strahler system - 1st through 12th Order by Strahler's (1952)

- Stream ordering is a quantitative method for describing the hierarchy of a drainage network.
- First-order streams are headwater channels with no upstream tributaries of the same order.
- When two streams of the same order join, the downstream segment increases by one order.
- When different orders join, the downstream segment retains the higher order.
- Strahler's system is widely used for drainage-network analysis.



3. Water Budget & Rainfall–Runoff

Tracking inflows, outflows and storage

How Water Moves Through a Watershed?

01 **Headwaters**

Rainfall and snowmelt in the upper areas begin to flow downhill.

02 **Tributaries**

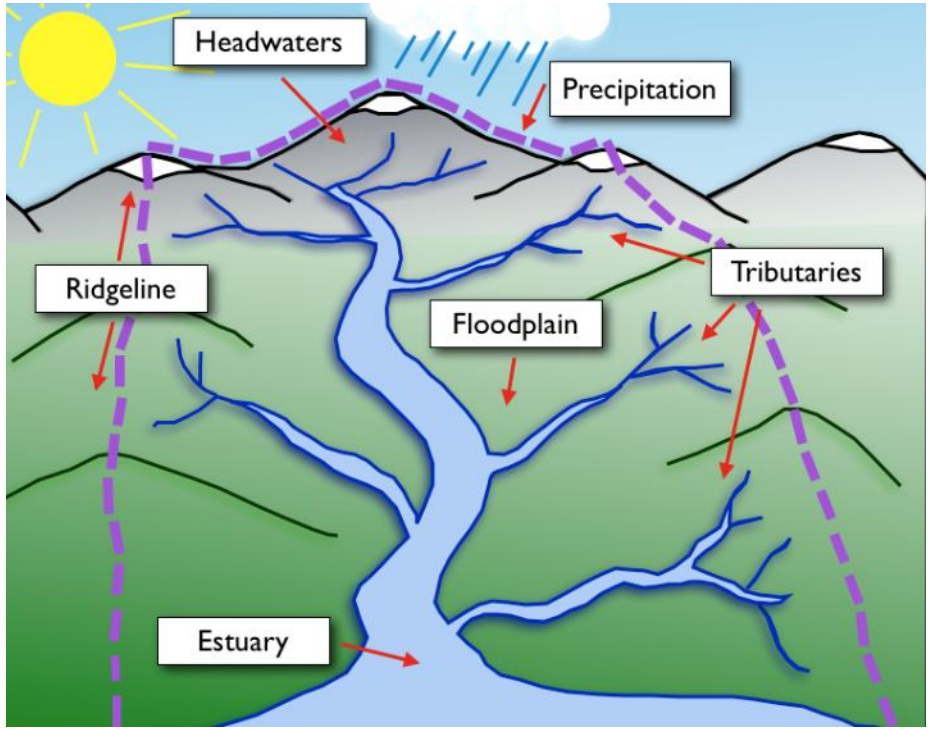
Small streams collect water from different parts of the watershed and join larger streams.

03 **Main Channels**

Water flows through the main river or stream, carrying water and sediment downstream.

04 **Outlet**

Water leaves the watershed through a common outlet, such as a river, lake, reservoir, or sea.



FROM MAP TO FIELD

Basin Water Balance

- Inflows: precipitation, surface-water inflow, groundwater inflow, and artificial imports.
- Outflows: evaporation, runoff, groundwater outflow, and artificial withdrawals.
- Storage: surface water, soil moisture, groundwater, snow/ice, and temporary/intercepted storage.
- Hydrologic mass balance: Inflow = Outflow ± change in storage.

$$\Sigma \text{ Inflows} - \Sigma \text{ Outflows} = \Delta \text{ Storage}$$

Rainfall–Runoff Response

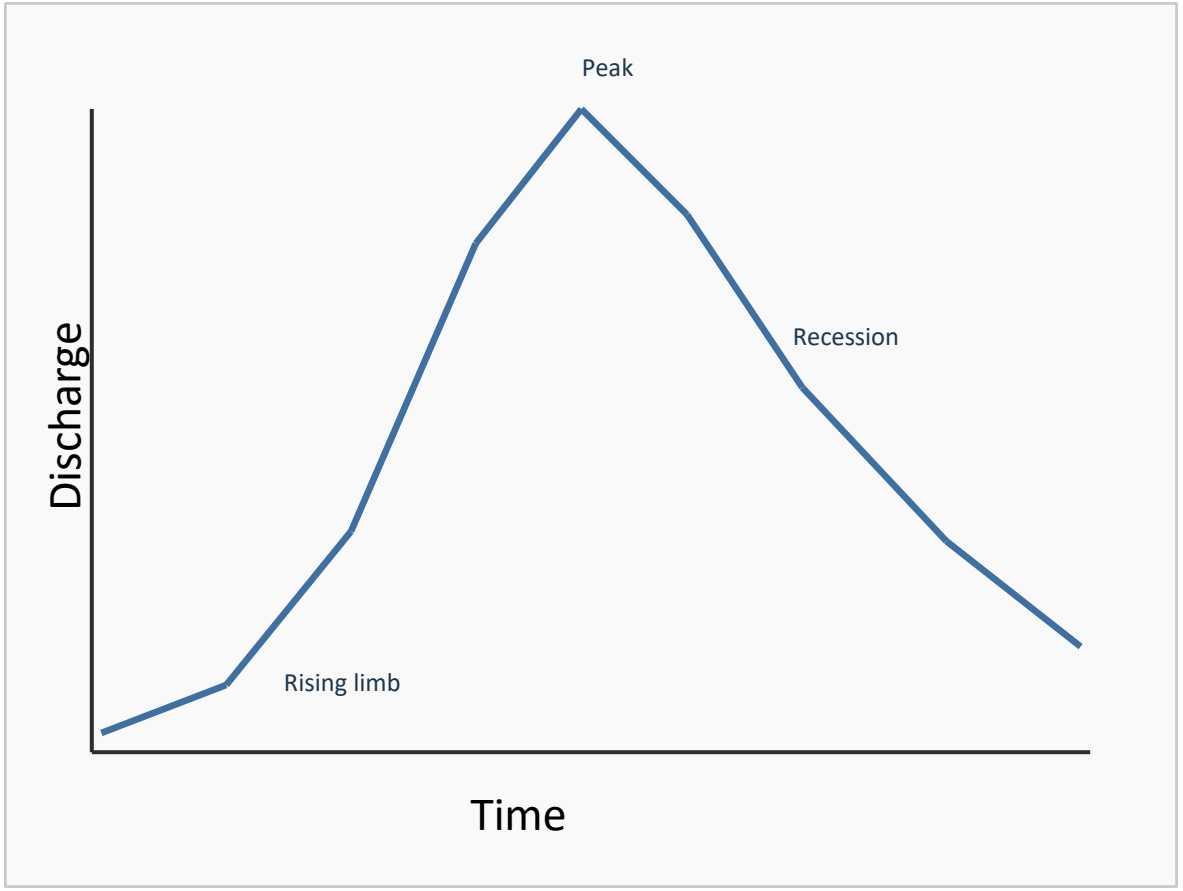
- A catchment can be treated as a system that transforms rainfall input into streamflow output.
- Inputs include rainfall amount, intensity, duration, and spatial distribution.
- The catchment controls the transformation through soils, land cover, slope, drainage, and storage.
- The observed output is discharge at the catchment outlet.

4. Hydrograph Analysis

From observed rainfall to streamflow

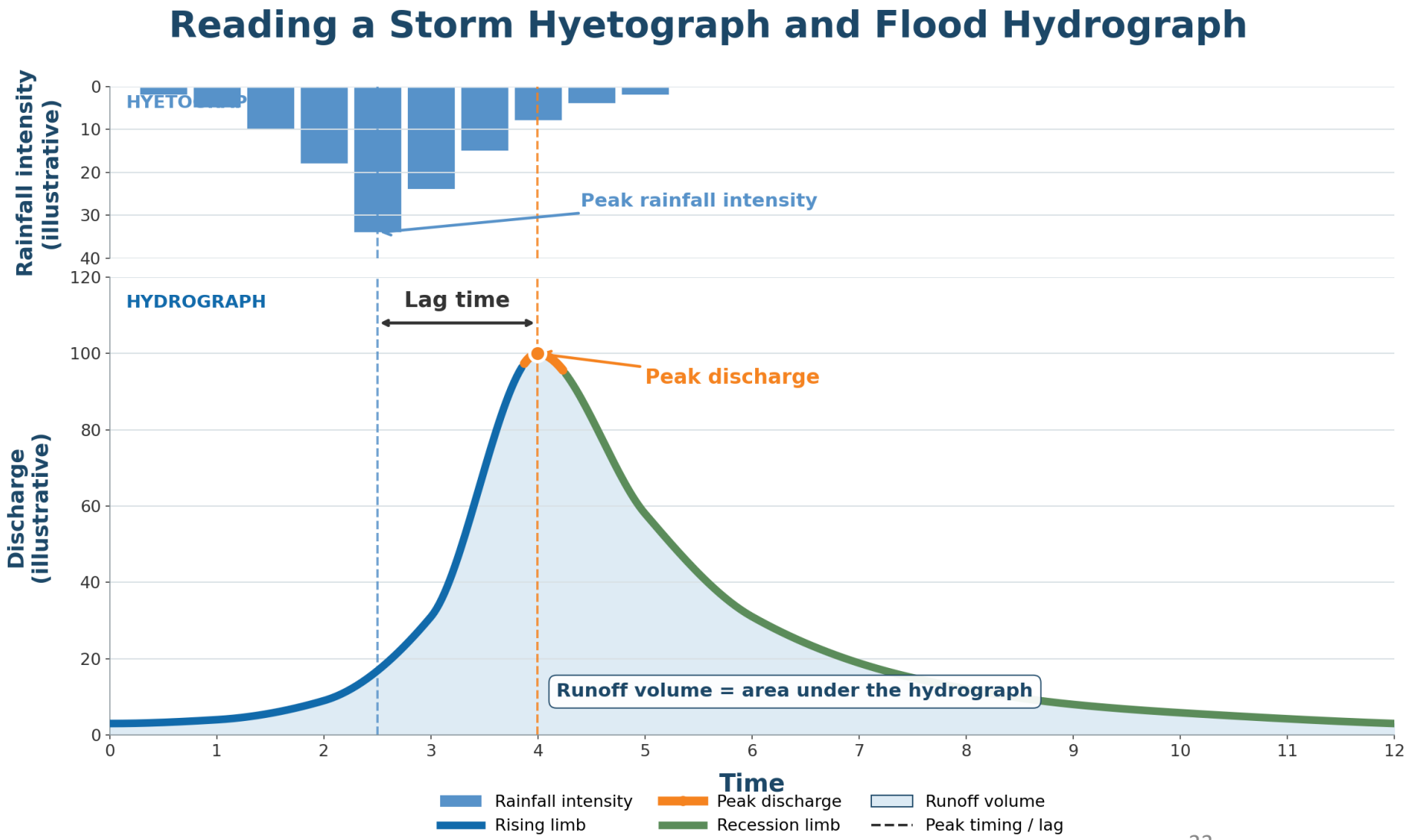
Hydrograph

- **Hydrograph:** discharge plotted against time at a channel location.



How do we read a Flood Hydrograph?

Peak, timing, and volume—all matter for design.

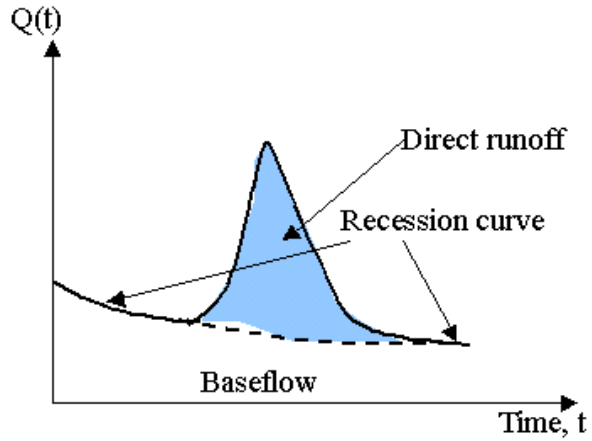


Hydrograph Analysis

- Hydrologic response of rainfall at the outlet of an area
- Hydrograph: Graph of discharge (cubic feet per second) in a channel vs. time
- Area under curve yields the volume of runoff

Stream flow = Direct Runoff + Base flow

Direct Runoff (DRO) = Rainfall Excess or (rainfall – losses)



Source: Prof. Ke-Sheng Cheng

What controls hydrograph shape?

Storm/climate

- Rainfall Intensity
- Rainfall Duration
- Rainfall Spatial Distribution
- Rainfall Temporal Distribution

Catchment

- Catchment Size and Shape
- Slope and Elevation
- Soil and Land cover
- Drainage density, Roughness, and Storage
- Antecedent Moisture

How Catchment Characteristics Change Response?

- Steeper slopes generally produce faster runoff and a sharper hydrograph response.
- Impervious or highly developed surfaces can reduce infiltration and increase rapid runoff.
- Larger catchments generally produce greater runoff volume for the same rainfall depth, but their response is often broader and slower.
- Vegetation, soil properties, channel length, drainage density, and natural/channel storage also modify the response.
- Snowmelt at high elevation can produce broader, lower-peaked flood responses than intense rainfall.

"Why Does the Same Rainfall Produce Different Runoff?



Watershed A	Watershed B
Gentle slopes	Steep slopes
Permeable soils	Shallow or less permeable soils
More vegetation	Sparse vegetation
More water soaks into the ground	More water flows over the surface
Less and slower runoff	**More and faster runoff**

Key message: Rainfall alone does not determine runoff.
Soil, slope, vegetation, and existing moisture conditions also matter.

Different watershed characteristics

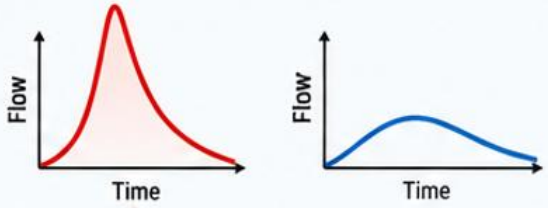
Same rainfall
(e.g. 50 mm)

Different runoff response

1. Slope (Topography)

Steeper slopes generate faster and higher runoff.



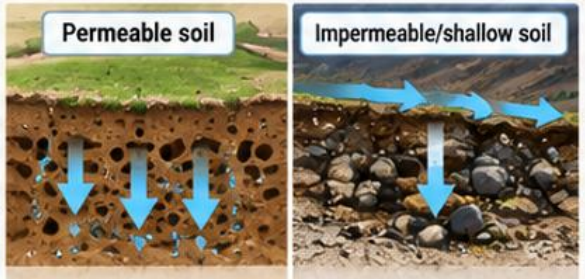


Higher and earlier peak

Lower and later peak

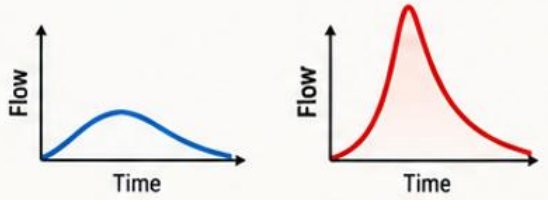
2. Soil and Infiltration

Permeable soils allow more water to soak in, reducing runoff.



More infiltration
Less runoff

Less infiltration
More runoff

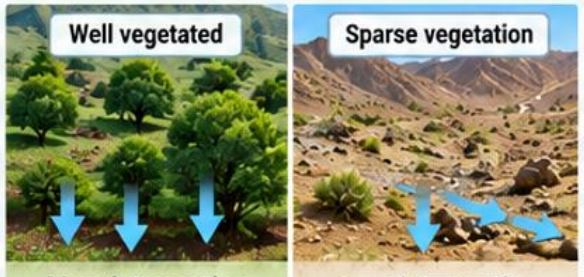


Lower peak

Higher peak

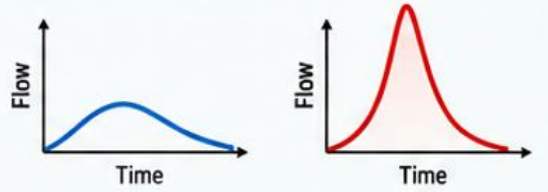
3. Vegetation/Land Cover

Vegetation slows runoff, increases infiltration and storage.



More interception and infiltration
Less runoff

Less interception
More runoff

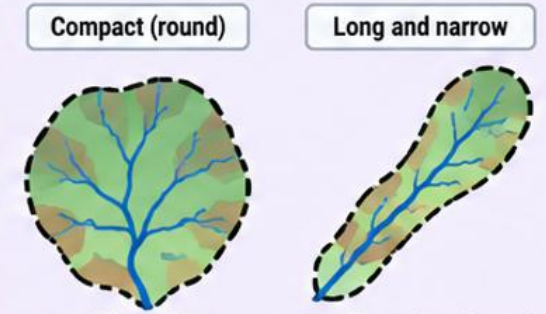


Lower and delayed peak

Higher and earlier peak

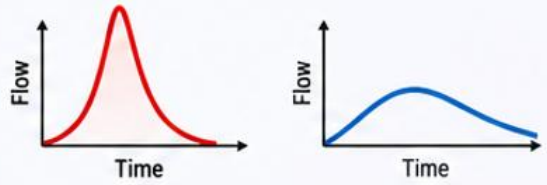
4. Watershed Shape

Watersheds with the same area can produce different hydrographs.



Runoff from all parts reaches the outlet quickly.

Runoff reaches the outlet at different times.

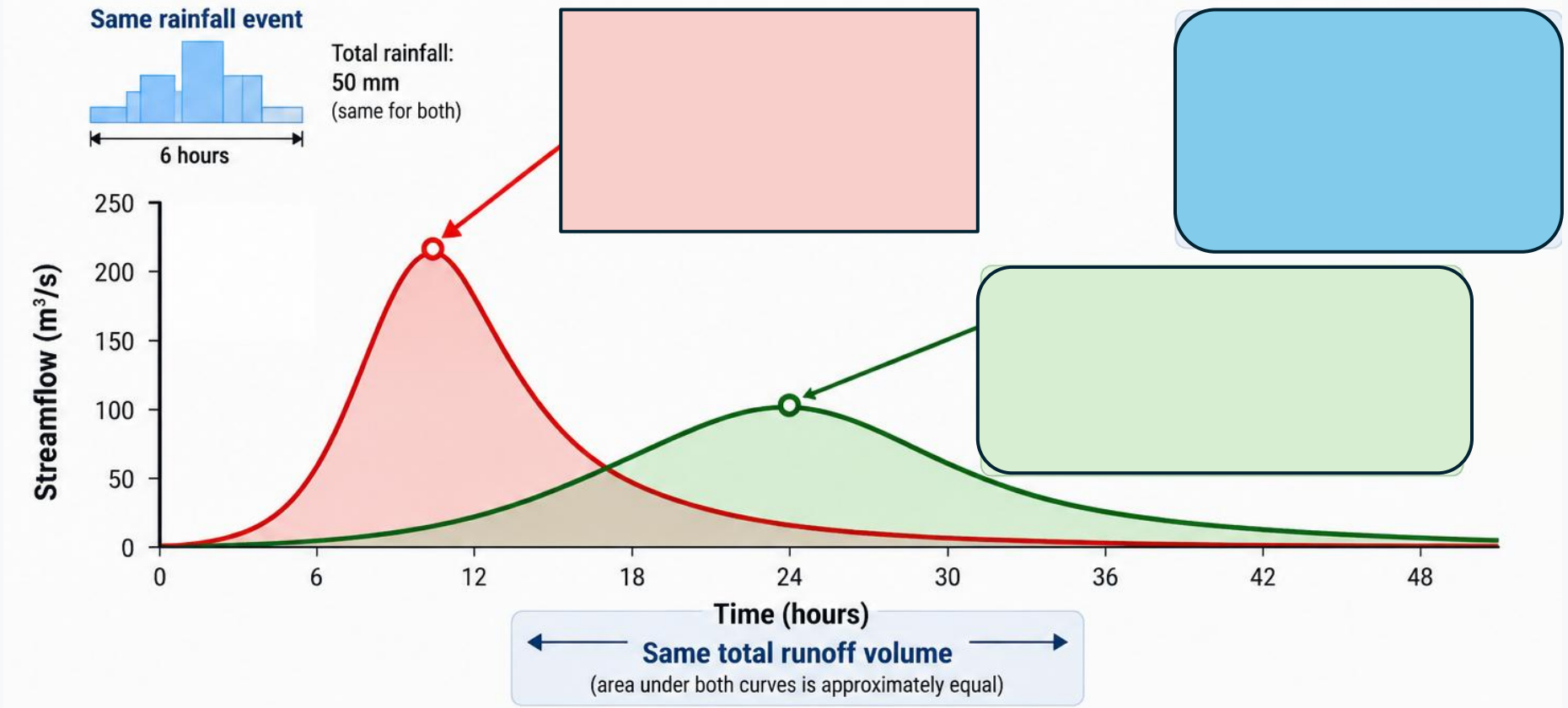


Higher and earlier peak

Lower and later peak

Same monthly rainfall—different runoff risk

The same rainfall event can produce different flood peaks and timing, even when the total runoff volume is similar.



05

Rainfall–Runoff Modeling

From observed rainfall and catchment characteristics to predicted discharge

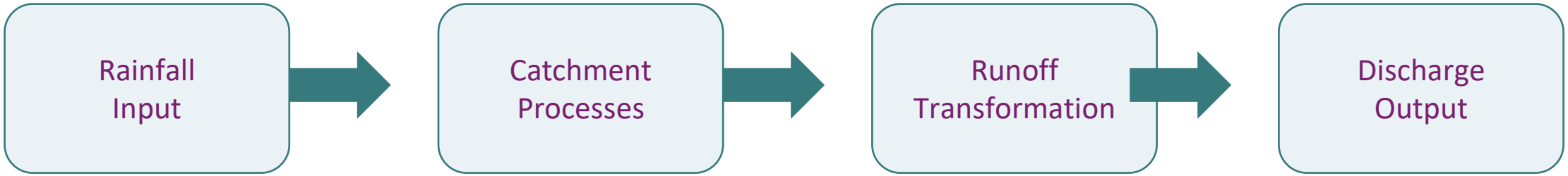
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Rainfall–Runoff Modeling Concept

Objective: estimate the runoff or streamflow response produced by a rainfall event or rainfall time series.

- Inputs may include rainfall, soil properties, land cover, topography, and antecedent conditions.
- Observed discharge is used for calibration and evaluation where measurements are available.
- Model choice depends on the purpose, data availability, catchment scale, and required level of detail.

A simple Conceptual Model



Losses and storage control how much rainfall becomes effective rainfall.

Catchment routing controls when that water reaches the outlet.

Gauge and Satellite data are Complements

Gauge strengths

Direct point measurement; long local records where maintained.

Gauge limitations

Sparse coverage; missing values; wind/exposure errors.

Satellite strengths

Spatially complete, near-real-time products.

Satellite limitations

Bias in mountains and intense localized storms; validate locally.

Choosing a rainfall–runoff method

The method should match the decision, watershed scale, and available data.

Rational Method: Estimates peak discharge for suitable small catchments; not event volume.

Curve Number Method: Estimates event runoff depth from rainfall, soil, and land-cover conditions.

Hydrological and Hydraulic (H&H) modeling: Can simulate runoff timing and routing when suitable data are available.

Use local evidence and check assumptions before accepting results.

Regression or ML as a Hydrologic Analysis Tool

- Regression can relate runoff or streamflow to explanatory variables when a suitable data record is available.
- Regression works best for long time periods.
- Possible predictors may include rainfall characteristics, basin descriptors, or other hydrometeorological variables.
- Regression is an empirical relationship; it should be evaluated using appropriate goodness-of-fit and validation measures.

Can we trust the model results?

A good-looking result is not enough without evidence.

Check rainfall depth, timing, and spatial coverage.

Compare simulated and observed flood peaks, timing, and volume.

Test reasonable ranges of uncertain parameters.

Validate with another event or independent field evidence.

Report uncertainty rather than relying on one number.

Key takeaways

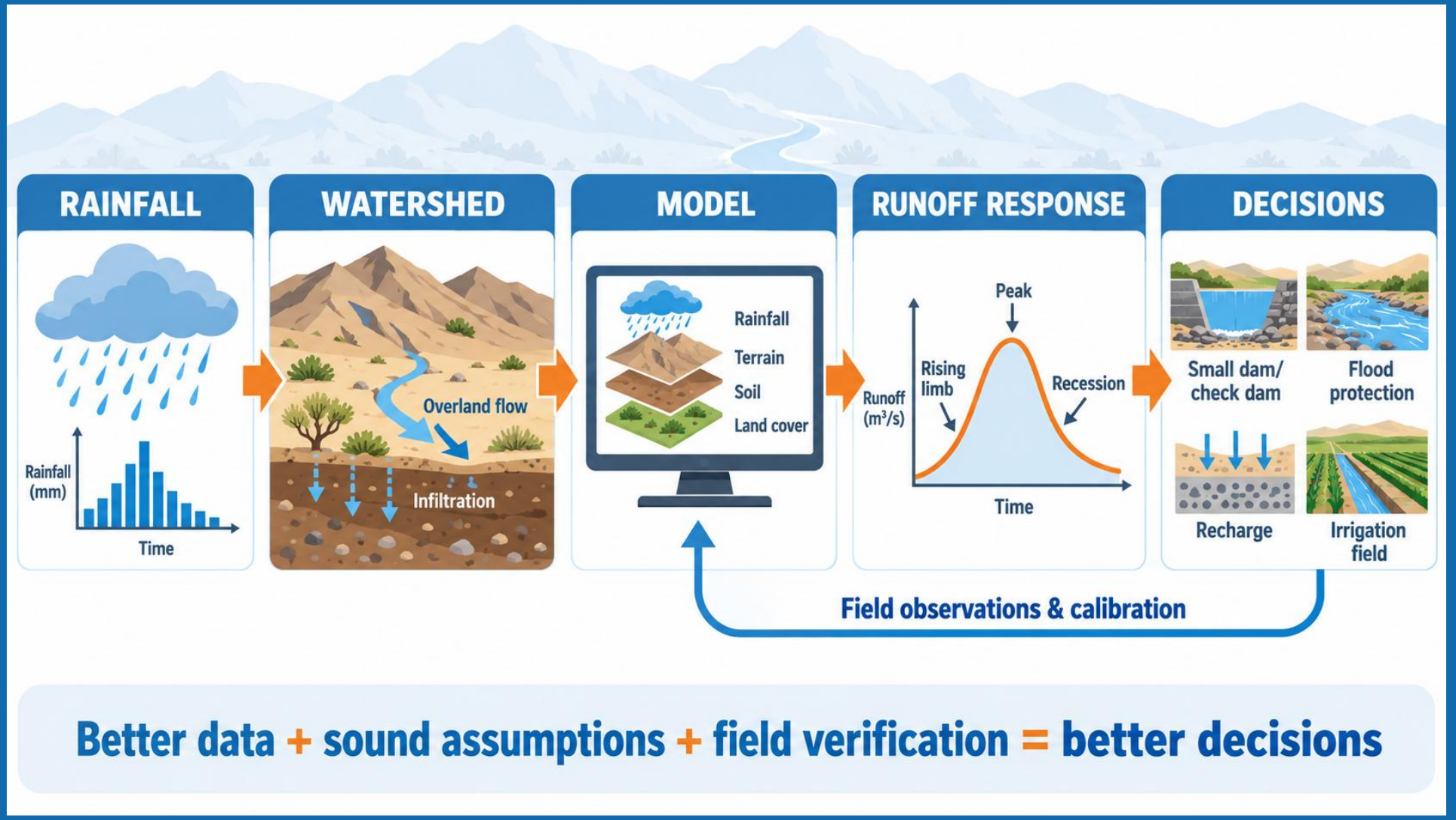
- Rainfall is the primary atmospheric input, but only part of it becomes effective rainfall and runoff.
- The watershed provides the spatial framework for hydrologic analysis.
- Topography controls drainage direction and watershed boundaries.
- The hydrograph shape reflects both storm and catchment characteristics.
- Rainfall–runoff modeling represents the transformation from rainfall input to discharge output.
- Good modeling requires appropriate data, clear assumptions, calibration, and validation.

Quiz: rainfall–runoff and watershed assessment

1. Why can two watersheds receiving the same rainfall produce different runoff?

2. What does the peak of a hydrograph represent?

3. Why should a DEM-derived watershed boundary be checked in the field?


RAINFALL
Rainfall (mm)
Time
WATERSHED
Overland flow
Infiltration
MODEL
Rainfall
Terrain
Soil
Land cover
RUNOFF RESPONSE
Peak
Rising limb
Recession
Runoff (m³/s)
Time
DECISIONS
Small dam/ check dam
Flood protection
Recharge
Irrigation field
Field observations & calibration
Better data + sound assumptions + field verification = better decisions

Thank You